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The Effect of Methylphenidate Treatment on Oxidative Stress Levels in Children Diagnosed with Attention-Deficit/Hyperactivity Disorder: An Observational Study

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Abstract

Background: Oxidative stress (OS) has been implicated in the pathophysiology of attention-deficit/hyperactivity disorder (ADHD), yet findings—particularly regarding treatment-related changes—remain inconsistent. This study evaluated short-term differences in circulating oxidant/antioxidant markers after methylphenidate (MPH) treatment and explored whether baseline biomarkers predict clinically significant symptom improvement.

Methods: In this single-center, prospective observational cohort study, medication-naïve children aged 6–11 years diagnosed with DSM-5-TR ADHD were assessed at enrollment and after 3 months of conventional MPH treatment. Clinical severity was measured using the Conners' Parent Rating Scale–Revised: Short Form (CPRS-R:S) and Conners' Teacher Rating Scale–Revised: Short Form (CTRS-R:S). Serum biomarkers included total oxidant status (TOS), total antioxidant status (TAS), superoxide dismutase (SOD), malondialdehyde (MDA), oxidized LDL (Ox-LDL), and the OS index (OSI = TOS/TAS). Changes were tested with paired analyses; correlations were examined between percent changes in scale scores and biomarkers; and ROC analyses evaluated baseline biomarkers' ability to identify a >50% reduction in CPRS-R:S and CTRS-R:S total scores.

Results: Thirty-nine participants completed both assessments (SOD analyses: $n = 37$). Significant reductions were observed in CPRS-R:S and CTRS-R:S total and subscale scores (all $p < .001$). TOS, TAS, SOD, MDA, and Ox-LDL decreased significantly over 3 months, while OSI showed a nonsignificant downward trend ($p = 0.066$). No significant correlations emerged between percent symptom change and percent biomarker change. In ROC analyses, baseline MDA, Ox-LDL, and TOS significantly discriminated >50% CPRS-R:S improvement (AUC ≈ 0.71–0.74), whereas for >50% CTRS-R:S improvement, several biomarkers demonstrated good discrimination (AUC ≈ 0.82–0.85).

Conclusions: Three months of MPH treatment was associated with robust clinical improvement and parallel reductions in multiple oxidant and antioxidant biomarkers. Baseline OS markers may help identify children most likely to show marked symptom reduction, though findings should be interpreted cautiously, given the sample size.

Keywords: attention-deficit/hyperactivity disorder (ADHD), oxidative stress, methylphenidate, observational study, prepubertal

Introduction

Attention-deficit/hyperactivity disorder (ADHD) is a multifaceted neurodevelopmental disorder with childhood onset, distinguished by hyperactivity, inattention, and impulsivity, and it can negatively affect an individual's academic and social functioning across the lifespan (American Psychiatric Association, 2022). Globally, its prevalence among children and adolescents is estimated to range from approximately 7.2% to 8.0%, and it is observed about twice as often in boys as in girls (Ayano et al., 2023; Elia et al., 2008; Faraone et al., 2021). The etiology of ADHD is multifactorial, involving an interplay of genetic predisposition, neurobiological susceptibility, and environmental risk factors (Faraone et al., 2014; Thapar et al., 2013).

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Recent studies suggest that oxidative stress (OS) and neuroinflammation may play a critical role in the ADHD pathogenesis (Corona, 2020; Visternicu et al., 2024). Brain tissue is particularly susceptible to oxidative damage due to its high oxygen consumption, abundant lipid content, and relatively limited antioxidant defense mechanisms compared with other organs (Ceylan et al., 2012; Visternicu et al., 2024). OS, defined as a disruption in the balance between antioxidant and oxidant systems, can trigger neurological damage by facilitating lipid peroxidation in cell membranes, inducing DNA damage, and impairing neurotransmitter receptor function (Lopresti, 2015; Visternicu et al., 2024).

Findings about the relationship between OS and ADHD remain inconsistent in the current literature; while some studies report increased OS and reduced antioxidant capacity in individuals with ADHD, others have found no significant differences (Başay et al., 2025; Guney et al., 2015; Miniksar et al., 2023; Oztop et al., 2012; Visternicu et al., 2024). Specifically, antioxidant response has a complex nature. Some studies highlight that, in response to an increased oxidant burden, the body may upregulate antioxidant levels (e.g., total antioxidant status [TAS]) through compensatory mechanisms (Guney et al., 2015; Selek et al., 2012). This may indicate that, once oxidant levels return to normal, the body may, as an adaptive response, downregulate its defense mechanisms (Aydın et al., 2024; Başay et al., 2025). Although most clinical studies suggest that methylphenidate (MPH) restores redox balance by strengthening antioxidant defenses, experimental models also provide evidence that stimulants may paradoxically induce oxidative damage; therefore, findings regarding the effects of MPH are inconsistent (Foschiera et al., 2022; Garre-Morata et al., 2024; Martins et al., 2006; Sanches et al., 2025).

First-line pharmacological agents for ADHD, such as the psychostimulant MPH, improve symptoms by blocking norepinephrine and dopamine transporters (Shier et al., 2013; Spencer et al., 2005). However, the effects of MPH on oxidative mechanisms in human plasma and the neuroadaptive consequences of long-term use remain controversial. Some recent studies report that MPH treatment may restore disrupted redox balance by enhancing antioxidant defense mechanisms and, in parallel with symptomatic improvement, may reduce serum OS (Foschiera et al., 2022; Garre-Morata et al., 2024; Novo et al., 2023; Visternicu et al., 2024).

This study aims to fill a gap in the literature by analyzing the potential effects of MPH treatment on plasma oxidant/antioxidant parameters in treatment-naive children aged 6–11 years with ADHD and also by examining the relations between these parameters and clinical rating scales.

Methods

Study design and setting

This single-center, prospective cohort (observational) study investigated changes in oxidant/antioxidant parameters in children diagnosed with ADHD according to DSM-5-TR criteria by comparing pretreatment values with those obtained at the third month of MPH treatment. The study was conducted at the Child Psychiatry Outpatient Clinic of Bakirkoy Prof. Dr. Mazhar Osman Training and Research Hospital for Psychiatry and Neurological Diseases, Istanbul, Türkiye. Participants were recruited between August 2024 and January 2025 and followed for 3 months.

Participants and eligibility criteria

Children aged 6–11 years presenting to the outpatient clinic with a previous or new ADHD diagnosis and for whom MPH initiation was planned were enrolled. This prepubertal age range was selected to reduce developmental (puberty-related) variability in OS-related processes and thereby improve interpretability of biomarker changes. Families received study information, and the legal guardian provided a written consent; written assent was obtained from literate children, and verbal assent from nonliterate children following verbal information.

Exclusion criteria were autism spectrum disorder, intellectual disability, or having other neurodevelopmental disorders; hereditary/metabolic disorders, structural brain abnormalities, or progressive neurological disease; active infection, allergic disease, or any chronic medical condition; alcohol or psychoactive substance use; current use of any psychotropic or nonpsychotropic medication; and any prior use of psychotropic medication for any reason, including ADHD medication.

Eligibility criteria were shared with experienced child psychiatrists providing active clinical care; potentially eligible cases were referred and reevaluated by the principal investigator. Diagnoses and comorbidities were assessed using the Schedule for Affective Disorders and Schizophrenia for School-Age Children—Present and Lifetime Version, DSM-5 Turkish Adaptation (K-SADS-PL-DSM-5-T) (Kaufman et al., 1997; Unal et al., 2019). Families were informed that participation would not affect routine clinical care.

Clinical follow-up and treatment

As an observational study, treatment followed routine practice and clinical indications. Only MPH-containing preparations were used. Formulation (immediate-release, intermediate-release, and long-acting [OROS]) and dosing were determined by the treating physician based on weight, clinical status, and adverse effects, and were titrated during follow-up. Routine visits were scheduled every 4 weeks, with earlier visits when needed (e.g., adverse effects or family concerns). The researcher's contact information was provided for adverse or unexpected events.

Blood sampling and sample processing

Participants attended between 09:00 and 12:00 following 10–12 hours of fasting. Sociodemographic and clinical data were recorded, and ≥5 mL venous blood was collected into a yellow-top gel serum separator tube. Samples were rested at room temperature for 10–20 minutes and centrifuged at approximately 1000 × g for 20 minutes, and the serum was separated into Eppendorf tubes and stored at −80°C until analysis. The serum samples were transferred from −80°C to a +4°C environment 15 hours before the scheduled analysis. After 15 hours at +4°C, the samples were removed and analyzed. After blood collection, MPH treatment was prescribed by the treating physician at an appropriate dose and formulation.

Participant flow

The study was planned to conclude once 39 participants completed both baseline and month-3 assessments; withdrawals were replaced by new participants. Overall, 53 participants provided written informed consent and entered the study; 14 withdrew for various reasons, and recruitment continued until 39 completers were reached. Analyses were conducted in participants with complete baseline and measurements at the third month.

Serum assays

Frozen serum samples stored at −80°C were transferred to +4°C and kept for 15 hours before analysis. The five biomarkers (TAS, total oxidant status [TOS], malondialdehyde [MDA], superoxide dismutase [SOD], and oxidized LDL [Ox-LDL]) were measured simultaneously using ELISA kits from Bioassay Technology Laboratory: TAS (Cat. No. E4350Hu; Shanghai, China), TOS (Cat. No. E1599Hu; Shanghai, China), MDA (Cat. No. E1371Hu; Shanghai, China), SOD (Cat. No. E0918; Shanghai, China), and Ox-LDL (Cat. No. E1521Hu; Shanghai, China). Assays were performed in the Medical Biochemistry Laboratories of Bakirkoy Dr. Sadi Konuk Training and Research Hospital by a biochemistry specialist according to manufacturers' ELISA protocols. Plate washing and reading were conducted using an Allsheng Microplate Washer APW-200 and an Allsheng Microplate Reader AMR-100T (Hangzhou, China). OS index (OSI) was calculated as TOS/TAS. In brief, standards and samples were loaded (40 μL serum), specific antibody (10 μL) was added, plates were incubated at 37°C for 60 minutes, washed five times, substrate solutions A and B (50 μL each) were added, plates were incubated for 10 minutes at 37°C in the dark, and stop solution (50 μL) was added before reading at 450 nm within 10 minutes (see Images 1 and 2).

Data collection instruments

Informed consent for voluntary participation: Legal guardians were informed about study aims, procedures, potential risks, and voluntariness; written informed consent was obtained (signed by the legal guardian and the researcher). For pediatric participants, assent procedures followed the "Turkish Medicines and Medical Devices Agency" guideline on "Ethical Approaches in Clinical Research Conducted in the Pediatric Population"; written assent was obtained from literate children and verbal assent from nonliterate children.

Sociodemographic and clinical data form: A researcher-developed form recorded sociodemographic characteristics (e.g., age, sex, educational status, family income level) and clinical data, including height, weight, and body mass index (BMI).

Diagnostic assessment: To confirm ADHD and assess comorbidities, the K-SADS-PL-DSM-5-T was administered by the researcher (Kaufman et al., 1997; Unal et al., 2019).

ADHD symptom severity scales

Parents completed the Conners' Parent Rating Scale–Revised: Short Form (CPRS-R:S) and teachers completed the Conners' Teacher Rating Scale–Revised: Short Form (CTRS-R:S) (Conners, 1997; Kaner et al., 2013a; Kaner et al., 2013b). Items are rated 0–3, with higher scores indicating greater symptom severity. Scales were administered at baseline and repeated at month 3.

Control of confounding factors

To reduce confounding, we included only 6- to 11-year-old children with confirmed ADHD, no significant psychiatric comorbidity, and no prior psychotropic exposure (including ADHD pharmacotherapy). Blood sampling was postponed in cases of active infection or acute transient illness until recovery, and children with chronic medical conditions were excluded. Seasonal and subclinical influences on OS markers could not be entirely eliminated and may have modestly affected results.

Sample size calculation

Sample size was based on a similar study examining the pre- and posttreatment change in the ARES serum biomarker after MPH (Guney et al., 2015). Using ARES values from that study and assuming $r = 0.50$ between paired measurements, an a priori power analysis was conducted in G*Power 3.1 for a paired-samples t-test (means: difference between two dependent means, matched pairs) (Faul et al., 2007, 2009). With two-tailed alpha = 0.05 and power (1 − beta) = 0.80, the estimated effect size was Cohen's $d_z = 0.56$, yielding a minimum required sample size of $n = 27$. To increase power, we targeted 39 completers.

Statistical analysis

Analyses were conducted using Jamovi (Jamovi project, 2025; R Core Team, 2025) (retrieved from https://www.jamovi.org). Descriptive statistics (mean, standard deviation, median, and percentages) were summarized in tables or bar charts. Normality was assessed using Shapiro–Wilk tests (Ghasemi and Zahediasl, 2012; Mohd Razali and Bee Wah, 2011). For paired comparisons, paired-samples t-tests were used when difference scores were normally distributed; otherwise, Wilcoxon signed-rank tests were applied. Effect sizes were reported as Cohen's d for paired t-tests and rank-biserial correlation for Wilcoxon tests (Cohen, 2013; Pallant, 2020). Correlations between percentage changes in total Conners' scores (CPRS-R:S and CTRS-R:S) (Percent change = [Baseline − Month 3]/Baseline × 100) and percentage changes in OS biomarkers were examined using Pearson correlations for normally distributed variables and Spearman's rho when appropriate. Exploratory (hypothesis-generating) ROC curve analyses evaluated whether baseline biomarkers discriminated participants achieving a ≥50% reduction in total CPRS-R:S and CTRS-R:S scores at month 3 (Coghill et al., 2020; Pelsser et al., 2009). For ROC analyses, AUC, 95% confidence interval (CI), p values, and optimal cutoffs (Youden index) with sensitivity/specificity were reported (Robin et al., 2010). Given the small sample size, CIs are reported, and ROC findings should be interpreted as exploratory. Two-tailed $p < 0.05$ was considered statistically significant. Benjamini–Hochberg false discovery rate (FDR-BH) correction was applied for multiple comparisons; this is indicated in the relevant tables/text.

Recruitment continued until 39 completers were reached, and analyses included participants with complete baseline and month-3 measurements. For SOD, two samples showed no reaction (no color change) and could not be read; thus, SOD analyses were based on $n = 37$.

Results

Participant flow diagram

A total of 53 participants were enrolled, 14 were lost to follow-up, and 39 completed the study. Reasons for withdrawal/exclusion ($n = 14$) were nonattendance and unreachable by phone ($n = 5$), children's refusal to take medication ($n = 2$), parental refusal due to stigma concerns ($n = 2$), parental concerns about potential adverse effects ($n = 2$), request for a second specialist opinion ($n = 2$), and dizziness at treatment initiation ($n = 1$). Participants who withdrew were replaced with additional eligible participants during the study period (Fig. 1). No enrolled participant (including those who later withdrew) reported rare or very rare adverse effects, and no serious medically significant adverse events occurred.

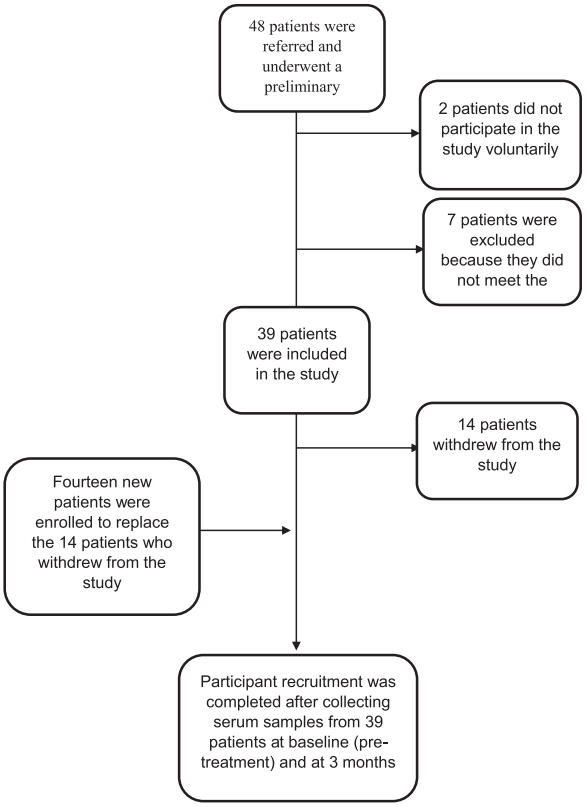


FIG. 1. Patient enrollment flowchart.

Sociodemographic and clinical characteristics

The mean age was 8.38 ± 1.11 years, and 67.7% were male. School grades ranged from preschool to seventh grade, with the majority of patients being in second grade (25.0%) and first grade (20.0%). Most parents were married (82.1%), and a first-degree family history of psychiatric disorders was reported in 23.1%. Academic achievement was classified as low (42.1%), moderate (36.8%), or high (21.1%); the preschool participant was excluded from this analysis (Table 1).

Changes in rating scale scores and serum parameters with treatment

Over 3 months of MPH treatment, symptom severity decreased substantially on both parent- and teacher-rated Conners' scales. Total CPRS-R:S and CTRS-R:S scores both showed significant improvement from baseline to month 3 (both $p < 0.001$), and improvements were observed across all subscales (all $p < 0.001$) (Table 2).

Serum OS parameters also changed over the treatment period (Table 3). TOS, TAS, SOD, MDA, and Ox-LDL decreased significantly from baseline to month 3 ($p \leq 0.021$ for all), whereas OSI showed a small, nonsignificant change ($p = 0.066$). After applying the Benjamini–Hochberg FDR correction for multiple comparisons, the significance status of the parameters did not change (Table 3).

Correlations between percentage changes in Conners' total scores (CPRS-R:S and CTRS-R:S) and percentage changes in OS biomarkers were not statistically significant (Table 4).

Exploratory ROC analyses examined whether baseline OS biomarkers discriminated participants achieving a ≥50% reduction in Conners' total scores at month 3. For CPRS-R:S responders ($n = 33$), baseline MDA, Ox-LDL, and TOS demonstrated significant discrimination with moderate performance (Fig. 2A; Supplementary Data). For CTRS-R:S responders ($n = 31$), discrimination was significant for multiple biomarkers and was generally stronger for teacher-rated outcomes (Fig. 2B; Supplementary Data). Detailed ROC parameters (cutoffs, sensitivity, specificity, and CIs) are provided in the Supplementary Data.

Discussion

In this prospective, observational, single-center cohort study, OS and clinical change after 3 months of MPH monotherapy in 6- to 11-year-old children with ADHD diagnosed according to DSM-5-TR were evaluated. Clinically, CPRS-R:S and CTRS-R:S total and subscale scores decreased significantly at the third month, consistent with current evidence supporting MPH efficacy in ADHD.

Biochemically, significant decreases were observed in TOS, MDA, and Ox-LDL, alongside reductions in TAS and SOD. OSI (TOS/TAS) showed a downward trend, but it was not statistically significant.

Table 1. Sociodemographic Characteristics of the Participants

Variable	n *(%)*	*Variable*	n *(%)*	Mean *(SD)*
Sex		Age (years)		8.38 (1.11)
Male	26 (67.7)	Grade level		
Female	13 (33.3)	Preschool	1 (2.6)	
Academic achievement[a]		1st grade	8 (20.0)	
Low	16 (42.1)	2nd grade	10 (25.0)	
Moderate	14 (36.8)	3rd grade	7 (17.9)	
High	8 (21.1)	4th grade	6 (15.4)	
Parental marital status		5th grade	4 (10.3)	
Married	32 (82.1)	6th grade	2 (5.1)	
Divorced	6 (15.4)	7th grade	1 (2.6)	
One parent deceased	1 (2.6)			
Psychiatric history in a first-degree relative				
No	30 (76.9)			
Yes	9 (23.1)			

Values are presented as *n* (%) unless otherwise indicated; age is reported as *Mean* (SD).
[a]One participant in preschool education was excluded from the academic achievement analysis.
SD, standard deviation.

Table 2. Comparison of Conners' Parent Rating Scale–Revised: Short Form and Conners' Teacher Rating Scale–Revised: Short Form Scores Between Baseline and 3-Month Follow-up

Scale/Subscale	*Baseline*	*3 months*	*Test statistic*	p	*Effect size*
CPRS-R:S Total	49.51 (15.94)	16.72 (10.65)	13.9[a]	**<0.001**	2.23
OPP	9.10 (4.32)	3.87 (4.03)	8.43[a]	**<0.001**	1.35
CPI	12.44 (4.57)	4.62 (3.79)	9.71[a]	**<0.001**	1.55
HYP	10.10 (4.92)	2.21 (2.51)	11.05[a]	**<0.001**	1.77
ADHD-I	24.21 (7.90)	8.13 (4.53)	13.71[a]	**<0.001**	2.20
CTRS-R:S total	38.64 (15.96)	15.87 (11.00)	12.68[a]	**<0.001**	2.03
OPP	4.00 (2.00–6.00)	0.00 (0.00–2.00)	666.00[b]	**<0.001**	1.00
CPI	6.18 (3.77)	3.77 (3.29)	7.54[a]	**<0.001**	1.21
HYP	9.90 (5.10)	3.51 (3.10)	9.54[a]	**<0.001**	1.53
ADHD-I–I	10.69 (4.35)	5.13 (3.09)	9.34[a]	**<0.001**	1.50
ADHD-I–H	7.03 (3.86)	2.18 (2.32)	9.01[a]	**<0.001**	1.44

The bold values were used to emphasize results that were statistically significant, based on the threshold of $p < 0.05$.

Values are presented as mean (SD) for normally distributed variables and median (Q1–Q3) for nonnormally distributed variables.

[a]Paired-samples *t*-test (Cohen's *d*).

[b]Wilcoxon signed-rank test (effect size = rank biserial correlation).

ADHD-I, Attention-Deficit/Hyperactivity Disorder Index; ADHD-I–H, ADHD Index–Hyperactivity; ADHD-I–I, ADHD Index–Inattention; CPI, cognitive problems/inattention; CPRS-R:S, Conners' Parent Rating Scale–Revised: Short Form; CTRS-R:S, Conners' Teacher Rating Scale–Revised: Short Form; HYP, hyperactivity; OPP, oppositional.

Previous studies primarily examine the relationship between OS and ADHD cross-sectionally and include ADHD and control groups. These studies generally report higher oxidant parameters in ADHD, whereas findings for antioxidant parameters are less consistent, although they are often lower (Corona, 2020; Joseph et al., 2015; Sezen et al., 2016). A meta-analysis likewise found higher oxidant biomarkers in ADHD but no difference in antioxidant biomarkers compared with controls; heterogeneity was particularly influenced by age, with larger effect sizes in children than in adults, leading the authors to recommend greater focus on pediatric samples (Joseph et al., 2015). Overall, studies on children and adolescents, particularly those targeting the preadolescent period, remain limited. As puberty involves substantial biological changes that may influence OS, we focused on a preadolescent sample to reduce potential confounders and to address this gap in the literature.

In studies assessing TOS, TAS, and OSI, oxidant markers such as TOS and OSI have generally been reported to be higher in patients with ADHD than in controls, whereas TAS has typically been lower (Koç et al., 2023). Similar patterns have been seen in several cross-sectional case-control studies (Kul et al., 2015; Sezen et al., 2016). In a 2015 study, TOS and OSI were higher in ADHD, while TAS did not differ between groups, and the authors attributed this to compensatory mechanisms (Guney et al., 2015). Findings have also been inconsistent across adulthood; one study reported no group differences in TOS or TAS (Aksoy et al., 2020), whereas another found higher TOS, OSI, and TAS in adults with ADHD, attributing elevated TAS as a compensatory response (Selek et al., 2012).

MDA is a metabolite of chain reactions that result from the oxidation of polyunsaturated fatty acids. Therefore, it serves as a marker of OS and has been linked to various neuropsychiatric disorders. In several studies conducted on ADHD, higher MDA levels have been reported in patients with ADHD compared with control groups (Bulut et al., 2013; Ceylan et al., 2010; Elhady et al., 2019).

For SOD, an antioxidant enzyme, the literature is inconsistent, similar to other antioxidant measures. Some studies have reported lower SOD levels in patients with ADHD (Ceylan et al., 2010; Lu et al., 2022). There are also studies reporting no significant difference in SOD levels between the ADHD and control groups (Ceylan et al., 2012; Russo, 2010).

Many of the aforementioned studies are cross-sectional case-control comparisons. Animal studies assessing the effects of

Table 3. Comparison of Serum Oxidative-Stress-Related Biomarkers Between Baseline and 3-Month Follow-up

Biomarker	*Baseline*	*3 months*	*Test statistic*	p	*Effect size*	p *(FDR–BH)*
TOS (U/mL)	6.30 (5.38)	5.63 (4.76)	4.06[a]	**<0.001**	0.65	**0.001**
TAS (U/mL)	7.79 (4.89)	7.33 (4.36)	2.42[a]	**0.021**	0.39	**0.025**
SOD (U/L)	175.89 (158.96)	160.89 (143.50)	3.21[a]	**0.003**	0.53	**0.004**
MDA (nmol/mL)	9.58 (7.20–15.15)	8.65 (6.96–14.42)	664.00[b]	**<0.001**	0.70	**<0.001**
Ox-LDL (ng/mL)	34.19 (26.35–61.58)	29.59 (25.03–63.50)	635.00[b]	**0.001**	0.63	**0.001**
OSI (TOS/TAS)	0.76 (0.12)	0.72 (0.12)	1.89[a]	0.066	0.30	0.066

The bold values were used to emphasize results that were statistically significant, based on the threshold of $p < 0.05$.

Values are reported as *Mean* (SD) for normally distributed variables and *Median (Q1–Q3)* for nonnormally distributed variables. Two samples were excluded from the SOD analysis due to failure to produce a color reaction (SOD, $n = 37$).

[a]Paired-samples *t*-test (effect size = Cohen's *d*).

[b]Wilcoxon signed-rank test (effect size = rank biserial correlation).

FDR-BH, False Discovery Rate—Benjamini–Hochberg procedure; MDA, malondialdehyde; OSI, Oxidative Stress Index; Ox-LDL, oxidized low-density lipoprotein; SOD, superoxide dismutase; TAS, total antioxidant status; TOS, total oxidant status.

Table 4. Correlation Between the Percentage Change in Conners' Scale Scores and the Percentage Changes in Blood Parameters

	CPRS-R:S total		*CTRS-R:S total*	
Variables	*r*	p	*r*	p
TOS (U/mL)	0.02[a]	0.896	–0.15[a]	0.375
TAS (U/mL)	–0.23[a]	0.156	–0.06[a]	0.738
SOD (U/L)	0.13[a]	0.439	0.06[a]	0.736
MDA (nmol/mL)	–0.07[a]	0.655	–0.06[a]	0.702
Ox-LDL (ng/mL)	0.03[a]	0.847	–0.17[a]	0.312
OSI (TOS/TAS)	0.03[b]	0.873	0.11[b]	0.488

Two samples were excluded from the SOD analysis due to failure to produce a color reaction (SOD, $n = 37$).

[a]Spearman.

[b]Pearson.

CPRS-R:S, Conners' Parent Rating Scale–Revised: Short Form; MDA, malondialdehyde; OSI, Oxidative Stress Index; Ox-LDL, oxidized low-density lipoprotein; SOD, superoxide dismutase; TAS, total antioxidant status; TOS, total oxidant status.

MPH on OS have resulted in highly inconsistent findings, and several explanations have been proposed. Acute, supratherapeutic exposure of MPH may induce OS in young rats due to insufficient antioxidant defenses, whereas chronic therapeutic exposure has been suggested to increase antioxidant enzyme activity and also adaptive mechanisms, thereby not enhancing OS in the brain (Sanches et al., 2023; Schmitz et al., 2012; Witt et al., 2010). Observational studies on treatment-related oxidative changes in humans remain limited. In the second phase of a 2015 study involving children aged 6–18 years, MPH was associated with increased antioxidant levels and reduced OSI (Guney et al., 2015). Similarly, another study in children and adolescents reported reductions in oxidant biomarkers alongside increases in antioxidant biomarkers during MPH treatment (Garre-Morata et al., 2024).

In this study, decreases in both oxidant and antioxidant levels after MPH treatment in children with ADHD were consistent with some reports but differed from others. Notably, our findings align with studies that emphasize compensatory mechanisms and interpret results within this framework. The literature recognizes that the oxidant and antioxidant systems operate in a complementary balance and influence one another. Recent evidence also suggests that low-dose chronic OS can enhance adaptive responses and thereby increase antioxidant activity (Wan et al., 2024; Wang and Wang, 2025). Antioxidants neutralize the body's oxidant burden. Therefore, for many antioxidant systems, oxidants that exert toxic effects serve as the substrates they act upon (Ji and Yeo, 2021). As the oxidant burden increases, it is predictable that antioxidant system activity may also increase. Indeed, when the aforementioned ADHD studies are taken into account, the observation that inconsistent findings predominantly involve antioxidant biomarkers suggests that antioxidant measures may vary in accordance with oxidant levels (Altun et al., 2018; Davinelli et al., 2025). Within this framework, our results suggest that the magnitude of change in oxidant biomarkers after MPH treatment was greater than that observed for antioxidant biomarkers. In other words, antioxidant measures showed a smaller decrease than oxidant measures. This pattern is consistent with a compensatory perspective. We interpret these findings as follows: When ADHD is treated with MPH, the oxidant burden decreases, and, as a result, there is less need for antioxidant defenses to detoxify it, leading to a secondary decline in antioxidant levels. Accordingly, OSI, an indicator of overall

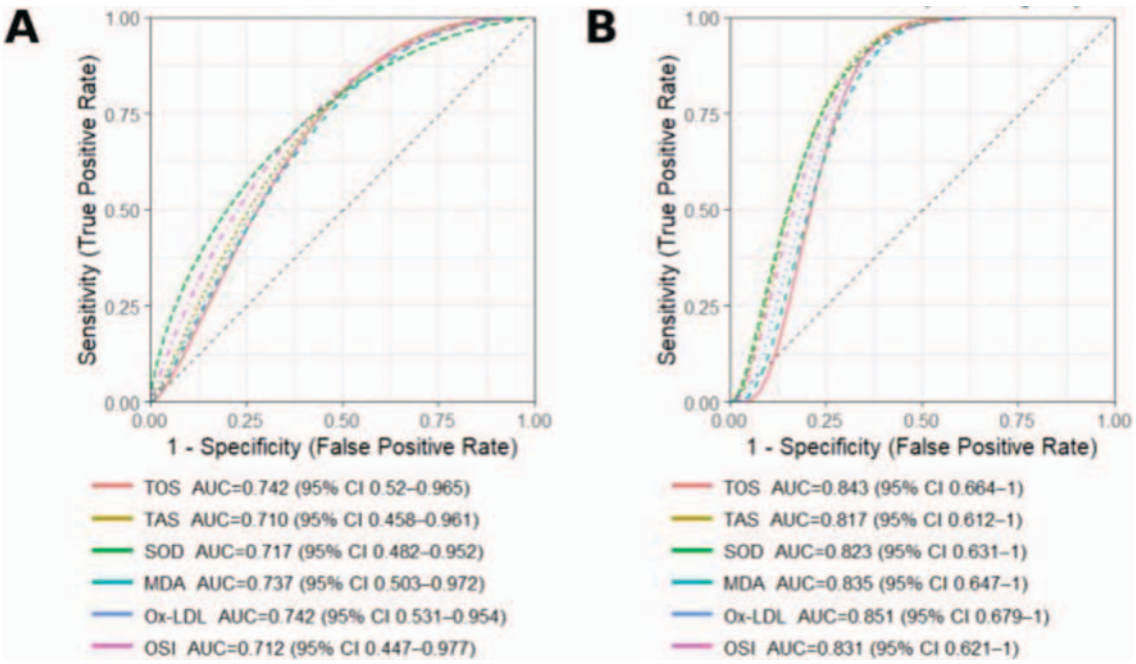


FIG. 2. Exploratory ROC curves for baseline oxidative stress biomarkers predicting ≥50% symptom reduction at month 3: **(A)** CPRS-R:S total score; **(B)** CTRS-R:S total score. AUC values with 95% confidence intervals are shown in the legend, and the diagonal dashed line indicates chance-level discrimination (AUC = 0.50). Given the limited sample size and the resulting stepwise appearance of empirical ROC curves, smoothed curve-fitting was applied to improve visual presentation. Full ROC statistics, including optimal cutoffs and corresponding sensitivity and specificity, are reported in the Supplementary Data. CPRS-R:S, Conners' Parent Rating Scale–Revised: Short Form; CTRS-R:S, Conners' Teacher Rating Scale–Revised: Short Form.

oxidative balance, showed a downward trend but narrowly missed statistical significance. We consider that this may also reflect compensatory processes. A slightly larger sample size might have been sufficient to detect a statistically significant change in OSI.

Another finding of our study was a significant reduction in Ox-LDL, an oxidant-related parameter, following MPH treatment. The cardiovascular effects of MPH have long been debated in the literature. Although it is generally considered safe, it is also clear that it may have cardiovascular effects that are usually of limited clinical significance. The prevailing view is that, while MPH is safe overall from a cardiovascular perspective, it can increase heart rate and blood pressure, and greater caution is warranted in patients with existing cardiovascular disease or a family history of arrhythmia. This is because case-level reports have described sudden arrhythmias potentially triggered by MPH (Hennissen et al., 2017; Man et al., 2023). To the best of our knowledge, we did not identify any prior studies that directly examined the relationship between ADHD and Ox-LDL. However, the literature includes many studies investigating the association between Ox-LDL and cardiovascular events, and Ox-LDL has been underlined in numerous reports as an early OS marker of cardiovascular disease (Hong et al., 2023; Papadea et al., 2023; Yu et al., 2025). Therefore, we considered it essential to examine the effect of MPH on Ox-LDL, an OS marker relevant to the cardiovascular system. Our findings suggest that, although MPH may have specific cardiovascular effects, these effects may not be mediated through Ox-LDL. Instead, by reducing Ox-LDL, MPH may reduce the adverse impact of OS on cardiovascular risk, similar to its effects on other OS-related mechanisms.

When the relationship between clinical severity and OS biomarkers is considered, the literature generally reports that no significant association is identified between OS parameters and clinical symptoms in patient groups (Avcil et al., 2021; Ceylan et al., 2010; Guney et al., 2015; Selek et al., 2012). In our study, no statistically significant association was found between percentage changes in total scores on parent- and teacher-rated scales and percentage changes in oxidative biomarkers. In contrast, ROC analyses suggested that baseline OS levels could discriminate individuals who showed a greater than 50% reduction in rating scale scores, with acceptable performance. This finding was more pronounced for the teacher-reported measure, in which lower baseline biomarker levels appeared to predict an over 50% reduction in scores. One possible explanation is that teacher ratings may be more sensitive to change in structured, high-demand settings (i.e., classrooms), where attentional control and behavioral regulation are continuously challenged and benchmarked against same-age peers. In contrast, parent ratings may reflect a broader range of contexts and may be more influenced by situational variability at home.

However, due to the limited number of participants and the wide CIs, these results were reported with CIs and should be interpreted as exploratory. In particular, predictive performance estimates derived from ROC analyses can be unstable in small samples, especially when the number of responders and nonresponders is limited. Under these conditions, ROC curves commonly appear "stepped" because classification thresholds are determined by a small number of distinct observed values, and minor changes in the sample can lead to noticeable shifts in the curve. Moreover, when multiple biomarkers are screened in a pilot dataset without internal validation, AUC estimates may be optimistically biased (overfitting/selection effects). Accordingly, our ROC findings should be viewed as hypothesis-generating signals that require confirmation in larger, controlled cohorts with prespecified predictors and validation procedures.

We did not identify prior studies that comparably examined this issue within an ADHD sample. Nevertheless, studies in other psychiatric conditions, such as depression and psychosis, have shown that lower levels of inflammatory biomarkers may predict better treatment response (Lindqvist et al., 2017; Mondelli et al., 2015; Zhang et al., 2019).

Conclusions

In this prospective, single-center observational cohort study, we examined changes in clinical symptoms over 3 months and OS biomarkers in 6- to 11-year-old children with ADHD diagnosed according to DSM-5-TR who received MPH monotherapy. At month 3, both parent- and teacher-rated symptom scales decreased significantly, indicating clinical improvement. Biochemically, OS-related markers also changed: Indices of oxidant burden declined, and antioxidant system measures decreased, with magnitudes smaller than the reductions in oxidant markers.

Although percentage changes in symptom ratings were not linearly associated with percentage changes in oxidative biomarkers, ROC analyses suggested that baseline OS levels may help identify children meeting a predefined response threshold (≥50% reduction in scale scores). Overall, these findings indicate that redox balance dynamics may shift during MPH treatment and that oxidative biomarkers merit further evaluation as potential biological correlates of treatment response.

Clinical Significance

This study corroborated the expected short-term clinical effectiveness of MPH for ADHD symptoms over 3 months using both teacher and parent ratings, while also demonstrating concomitant decreases in serum OS markers. Notably, antioxidant measures that counteract the oxidant burden also declined, which may reflect an adaptive downregulation in parallel with reduced oxidant load during treatment.

Because most prior work is cross-sectional and prospective studies often include mixed child–adolescent samples, focusing on a clinical cohort aged 6–11 years adds developmentally relevant evidence for ADHD. Although OS biomarkers are not currently recommended for routine clinical use, the ROC results suggest that baseline biomarker levels may help differentiate marked responders. These findings are exploratory and could guide future research on personalized monitoring and response prediction. Given the observational design, causal inference is limited, and these results should be interpreted as exploratory until replicated in larger controlled studies. Future studies should incorporate matched healthy controls and more granular stress-related phenotyping to clarify the specificity and clinical utility of these biomarkers. Because the study lacked a placebo or healthy control group, we cannot fully exclude time effects, regression to the mean, or nonspecific influences on OS biomarkers; therefore, biomarker changes should not be interpreted as causal effects of MPH.

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Authors' Contributions

E.F.A.: Conceptualization, data curation, methodology, formal analysis, investigation, and writing—original draft. A.A.: Methodology, supervision, and writing—original draft. G.K.: Methodology, project administration, supervision, and writing—review and editing. H.E.I.: Methodology, resources, and writing—review and editing. B.K.O.: Methodology and writing—review and editing. I.E.G.F.: Investigation and writing—review and editing.

Ethical Considerations

Ethical approval was obtained from the Republic of Türkiye Ministry of Health, Bakirkoy Dr. Sadi Konuk Training and Research Hospital Clinical Research Ethics Committee on March 18, 2024 (Protocol code: 2024/111; Decision No: 2024-05-27).

Consent to Participate

Written informed consent was obtained from the parents or legal guardians of all participating children after the consent form was reviewed with them. The study was also explained to the children in an age-appropriate manner; written assent was obtained from children who were able to read and write, and verbal assent was obtained from those who were not.

Consent for Publication

Written permission for publication was obtained from the patients and their parents or legal guardians, confirming that the data could be used for scientific purposes without disclosing identities or compromising confidentiality.

Data Availability

The data and materials supporting the conclusions of this article are publicly available in Zenodo (DOI: 10.5281/zenodo.18284252): https://doi.org/10.5281/zenodo.18284252 (Altunkılıç, 2026).

Author Disclosure Statement

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Supplementary Material

Supplementary Data

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